DENTAL AI PIPELINE

Automated OPG Analysis — Caries & Bone Loss Detection

Full Project Report | Generated: April 03, 2026

Project Type	Medical AI — Dental Radiology
Input	OPG (Orthopantomogram) X-ray
Output	Structured per-tooth dental report
Report Format	Tooth [FDI] → Caries + Bone Loss
Phase 1 Status	COMPLETE — Caries Detection (ResNet18)
Phase 2 Status	COMPLETE — Caries Severity (ResNet18)
Phase 3 Status	IN PROGRESS — Tooth Detection (YOLO)
Phase 4 Status	IN PROGRESS — Bone Loss (Dataset Explored)

1. PROJECT OVERVIEW

1.1 Goal

Build a fully automated system that takes a single OPG X-ray as input and produces a structured per-tooth dental report identifying caries and bone loss for every affected tooth using FDI numbering.

1.2 Final Output Format

The system outputs a structured report containing only affected teeth:

```
Tooth [FDI number]
→ Caries: Yes/No + Severity (mild / moderate / severe)
→ Bone loss: Severity (mild / moderate / severe)
```

1.3 Full System Pipeline

Stage	Component	Input	Output
1	Tooth Detection	Full OPG	Tooth bboxes + FDI index
2	Tight Crop	Tooth bbox x1.0	224x224 crown crop
3	Caries Model	Tight crop	Caries Yes/No + Severity
4	Expanded Crop	Tooth bbox x1.8	224x224 bone-visible crop
5	Bone Loss Model	Expanded crop	Bone loss Severity
6	Filter + Report	All tooth outputs	Final structured report

Critical design constraint: tight crops for caries (crown focus) vs expanded crops for bone loss (must include surrounding alveolar bone). Using the wrong crop type will cause model failure.

1.4 Crop Strategy

Model	Crop Factor	Region	Why
Caries (ResNet18)	1.0x (tight)	Crown only	Caries appears on enamel/dentin
Bone Loss (ResNet18)	1.5x W / 2.0x H	Crown + root + bone	Bone level visible below root

Asymmetric expansion (more vertical than horizontal) is required because teeth in OPG have mean aspect ratio 0.35 — they are 3x taller than wide. Horizontal over-expansion bleeds into adjacent teeth.

2. COMPLETED PHASES

2.1 Phase 1 — Caries Detection

Status: COMPLETE

- Architecture: ResNet18 pretrained on ImageNet, fine-tuned on tooth crops
- Input: 224x224 tight crop of crown region
- Output: Binary — Caries Yes / No
- Training: Class-balanced dataset with weighted loss
- Normalization: ImageNet mean/std (must match bone loss pipeline)

2.2 Phase 2 — Caries Severity

Status: COMPLETE

- Architecture: ResNet18 (same backbone as Phase 1)
- Input: 224x224 tight crop
- Output: 3-class — Mild / Moderate / Severe
- Optimization: Class-balanced, optimized for minority class recall
- Key decision: Severity only runs if Phase 1 detects caries — two-stage design

2.3 Phase 3 — Tooth Detection (YOLO)

Status: IN PROGRESS — handled separately

- Model: YOLOv8 or YOLOv11 for tooth bounding box detection
- Input: Full OPG (840x1615 px)
- Output: Per-tooth bboxes with tooth position index
- FDI assignment: Sort detected bboxes left-right, upper/lower jaw separation
- Training data source: Tufts teeth_bbox.json (26,005 tooth annotations, 968 images)
- Tooth numbering in Tufts: Universal system (1-32) — must convert to FDI

Universal to FDI conversion note: Universal tooth 1 = FDI 18 (upper right third molar). Mapping must be applied after YOLO detection for the final report to use correct FDI numbers.

3. PHASE 4 — BONE LOSS: DATA EXPLORATION

3.1 Dataset: Tufts Dental OPG

The Tufts Dental Dataset is a multi-annotator OPG radiograph collection with expert radiology reports, per-tooth bounding boxes, segmentation masks, and expert gaze tracking data. Full exploration was conducted before any modeling decisions were made.

3.2 Dataset Structure

Folder	Contents	Count
Radiographs/	OPG X-ray images	1000
Segmentation/maxillomandibular/	Full jaw region masks	1000
Segmentation/teeth_mask/	Per-tooth instance masks	308
Segmentation/teeth_bbox.json	Per-tooth bounding boxes	26,005 teeth
Segmentation/teeth_polygon.json	Per-tooth polygon annotations	1000 records
Expert/expert.json	Expert radiology report text	1000 records
Expert/mask/	Expert pathology region masks	439
Expert/gaze_map/	Expert gaze heatmaps	1000
Student/student.json	Student annotations	941 records

3.3 Image Quality Findings

Metric	Value	Implication
Image size	840 x 1615 px (uniform)	No resizing needed pre-crop
Mean pixel intensity	80-180 (normal dist)	Consistent exposure
Pixel std deviation	45-75	Good internal contrast
Michelson contrast	0.999 (mean)	High quality radiographs
Aspect ratio	All identical	No corrupted/rotated images

Conclusion: No image quality filtering required. Dataset is radiographically uniform.

3.4 Annotation Format Findings

expert.json structure:

- Format: List of 1000 records
- Join key: "External ID" field (e.g. "53.JPG") — NOT a dict key
- "Description": Free-text radiology diagnosis
- "Label.objects": List of polygon annotations per finding
- Bug caught: Initial parsing used dict keys — produced 0 coverage matches. Fixed by extracting stem from "External ID".

teeth_bbox.json structure:

- Format: List of 1000 records
- "External ID": filename join key
- "Label.objects": list of teeth, each with "title" (tooth number) and "bounding box"
- Coordinate format: [y1, x1, y2, x2] corner points — NOT standard x,y,w,h
- Bug caught: Assumed x,y,w,h format. Actual format required conversion: w=x2-x1, h=y2-y1, with x and y swapped.
- Tooth IDs: Universal numbering 1-32 plus deciduous A-T (pediatric cases present)

3.5 Mask Type Analysis

Mask Type	Pixel Values	Encoding	Use For
maxillomandibular/	0-18+	Jaw region blob (binary)	Jaw localization
teeth_mask/	0-18+	Per-tooth instance IDs	YOLO training data
Expert/mask/	0-42, 213-255	Per-lesion instance IDs	Pathology localization

Critical finding: Expert pathology masks do NOT encode bone loss severity. Each blob = one annotated lesion instance. Pixel value = instance ID, not class. This ruled out direct U-Net severity segmentation.

3.6 Expert Label Analysis

Category	Count	Percentage
Normal (within normal limits)	660	66.0%
Pathological finding (any)	340	34.0%
Total	**1000**	**100%**

Labels are free-text radiology reports, not structured classes. Severity levels (mild/moderate/severe) are mentioned in text but are not available as structured labels. Label extraction requires NLP keyword mapping.

3.7 Corrected Coverage Matrix

Initial coverage matrix showed 0 matches due to ID format mismatch bug. After fix, corrected numbers:

Intersection	Count	Notes
Radiographs (total)	1000	Baseline
Radio + Expert labels	1000	Full coverage
Radio + Bbox	968	32 images missing bbox
Radio + Teeth mask	308	Partial — instance seg subset
Radio + Expert pathology mask	439	Pathology annotation subset
Radio + Bbox + Expert label	968	Core training set
Radio + Bbox + Expert mask	423	Full annotation overlap
FULL: Radio+Bbox+Expert+ExpMask	**423**	**Working training pool**

3.8 Tooth Bounding Box Statistics

Metric	Value	Pipeline Implication
Total tooth bbox records	26,005	Rich per-tooth training data
Unique images with bboxes	968	Near-complete coverage
Avg teeth per image	26.9	Close to full dentition

Tooth width range	7-185 px (mean 63)	Varies significantly by tooth type
Tooth height range	13-350 px (mean 145)	Much taller than wide
Aspect ratio (w/h)	Peaks at 0.35-0.40	Teeth are ~3x taller than wide
Degenerate bboxes (w or h = 1px)	102 records	Filter before training
Deciduous teeth (A-T)	Present	Pediatric cases in dataset

3.9 Expert Mask Deep Analysis

Metric	Value
Total expert mask files	439
Empty masks (all zero = normal)	285
Masks with annotations (pathological)	154
Also have bbox annotation	423
Pixel value range (non-zero)	1-42 (low) and 213-255 (high)

Alignment confirmed: all 285 empty masks correspond to "Within normal limits" descriptions. All 154 annotated masks correspond to specific pathology descriptions. Perfect label-mask consistency.

3.10 Bugs Caught and Fixed

Bug 1 — JSON ID mismatch

Coverage matrix showed 0 matches. Root cause: code compared filename stems against dict keys. Actual join field is nested 'External ID' inside each record. Fix: extract Path(item['External ID']).stem as join key.

Bug 2 — Bbox coordinate format

Assumed standard [x, y, w, h]. Actual format: [y1, x1, y2, x2] corner points. Required: w = x2-x1, h = y2-y1, x and y swapped. Without fix, all crops would have been extracted from wrong image locations.

Bug 3 — Mask encoding assumption

Assumed expert masks encode severity classes. Actual encoding: per-lesion instance IDs. This ruled out direct U-Net severity segmentation and changed the architecture decision from segmentation to classification.

Bug 4 — PIL DecompressionBombError

PIL rejected large composite figures as security threats. Fix: PILImage.MAX_IMAGE_PIXELS = None and resize figures before PDF embedding.

Bug 5 — Font 404 error

DejaVu font wget from GitHub returned 404 HTML page, not a TTF binary. Fix: apt-get install fonts-dejavu-core and copy from system font directory.

4. PIPELINE DECISION FROM EXPLORATION

4.1 Why U-Net Severity Segmentation is Not Viable

- Expert masks encode lesion instance IDs, not severity class labels
- No pixel-level mild/moderate/severe annotation exists anywhere in dataset
- 154 annotated samples is insufficient for U-Net training from scratch
- CEJ estimation from OPG crops requires ground truth CEJ points — not available
- Post-processing chain (crest extraction, apex detection, % bone loss) has 3 unresolved fragile points with no validation data to calibrate against

4.2 Decided Architecture: Binary ResNet18 Classifier

DECISION: ResNet18 binary classifier — bone loss present / absent

Property	Value
Architecture	ResNet18 pretrained on ImageNet
Input	224x224 expanded crop (1.5x W, 2.0x H)
Output	Binary — Bone loss Present / Absent
Training set	423 images (full annotation overlap)
Positive class	154 images (pathological findings)
Negative class	285 images (normal / empty mask)
Imbalance ratio	1.85x (MODERATE)
Imbalance handling	Class-weighted loss
Normalization	ImageNet mean/std (same as caries pipeline)
Label source	Expert mask empty/non-empty + description keywords
Evaluation metric	Per-class F1 (target >= 0.70 for both classes)

4.3 Label Assignment Strategy

The dataset has image-level labels, not per-tooth labels. Label assignment to individual tooth crops follows this logic:

Condition	Crop Label
Expert mask is empty AND description = "within normal limits"	NEGATIVE — all tooth crops from this image
Expert mask has annotation AND description mentions pathology	POSITIVE — crops that spatially overlap with mask blob
Expert mask has annotation but crop does not overlap	NEGATIVE — this specific crop
No expert mask available for this image	EXCLUDE from training (uncertain label)

Spatial overlap strategy: expand tooth bbox by factor, check IoU with expert mask blob bounding rect. If IoU > 0.1, assign positive label. This is more principled than image-level label assignment to all crops.

5. NEXT STEPS — WHAT TO BUILD

5.1 Immediate Next Step: Dataset Construction

Before any model training, the processed dataset must be built on disk in the standard format. This is the most critical step — everything downstream depends on correct label assignment.

Step	Task	Output
1	Parse bbox JSON with correct coordinate conversion	df_bbox.csv with x,y,w,h
2	Parse expert.json with External ID join	df_expert.csv with stem + description
3	Load expert masks, classify empty vs annotated	df_joined.csv with mask_empty flag
4	For each annotated image: expand tooth bbox (1.5x W, 2.0x H)	Expanded crop coordinates
5	Compute IoU between expanded crop and expert mask bbox	Per-crop binary label
6	Add +/-15% bbox jitter to simulate YOLO noise	Augmented crop set
7	Save crops as 224x224 PNG files in class folders	processed_dataset_tufts/train/0/ and /1/
8	Stratified train/val split (80/20)	Split preserving class ratio
9	Verify: count files per split per class	Dataset ready for training

5.2 Dataset Directory Structure (Target)

```
processed_dataset_tufts/
train/
0/ (no bone loss) — ~228 images
1/ (bone loss) — ~123 images
val/
0/ (no bone loss) — ~57 images
1/ (bone loss) — ~31 images
```

5.3 Model Training

Component	Specification
Base model	ResNet18 pretrained (torchvision.models.resnet18)
Final layer	Replace FC layer: 512 → 2 (binary output)
Loss function	CrossEntropyLoss with class_weight=[1.85, 1.0]
Optimizer	Adam, lr=1e-4
Scheduler	ReduceLROnPlateau (patience=5)
Batch size	16
Epochs	50 (early stopping on val F1)
Augmentation	Horizontal flip, brightness jitter, rotation +-10 deg
Normalization	ImageNet mean=[0.485,0.456,0.406] std=[0.229,0.224,0.225]

Early stopping	Monitor val F1 macro, patience=10

5.4 Evaluation Criteria

Metric	Target	Why
Class 0 F1 (normal)	>= 0.75	Avoid false positives in report
Class 1 F1 (bone loss)	>= 0.70	Clinical sensitivity requirement
Macro F1	>= 0.72	Overall model quality
Accuracy	Secondary only	Misleading with imbalanced data
Confusion matrix	Must show	Identify which class fails

5.5 Integration with Full Pipeline

After bone loss model is trained and evaluated, it must integrate with the YOLO tooth detection and caries pipeline:

Integration Point	Requirement
YOLO output format	Bbox [x,y,w,h] + tooth position index
FDI assignment	Convert YOLO position index to FDI using left-right sort + jaw separation
Crop generation	Same expanded crop function used in training (1.5x W, 2.0x H)
Normalization	Identical to training: ImageNet mean/std
FDI metadata	Attach FDI number to every crop before inference
Report assembly	Join caries output + bone loss output on FDI key
Filter logic	Include tooth in report only if caries=YES or bone_loss=PRESENT

5.6 Universal to FDI Conversion Table

Tufts uses Universal numbering (1-32). Final report requires FDI. Conversion must be applied at the YOLO output stage:

Universal	FDI	Universal	FDI	Universal	FDI	Universal	FDI
1	18	2	17	3	16	4	15
5	14	6	13	7	12	8	11
9	21	10	22	11	23	12	24
13	25	14	26	15	27	16	28
17	38	18	37	19	36	20	35
21	34	22	33	23	43	24	44
25	45	26	46	27	47	28	48
29	31	30	32	31	33	32	34

Note: Standard Universal-to-FDI mapping. Verify against clinical reference before deployment.

6. KEY RISKS AND MITIGATIONS

Risk	Level	Mitigation
Small training set (423 images)	HIGH	Heavy augmentation (flip, brightness, rotation, gaussian noise). Transfer learning
Image-level labels → noisy per-crop assignment	MEDIUM	Use spatial IoU with expert mask for label assignment instead of assigning image
YOLO crop variability at inference	MEDIUM	Add +/-15% bbox jitter during training to simulate YOLO noise. Model must be ro
Moderate class imbalance (1.85x)	LOW-MEDIUM	Weighted loss function. Stratified split. Monitor per-class F1 — do not report agg
Universal vs FDI numbering mismatch	HIGH	Apply conversion table at YOLO output stage. Validate by overlaying FDI numbe
Deciduous teeth (A-T) in training data	LOW	Filter out deciduous tooth crops (IDs A-T) before training. Adult dentition only for
Domain shift: Tufts vs real deployment OPGs	MEDIUM	Tufts images are 840x1615 uniform. Real OPGs may vary. Apply CLAHE norma

7. CONTEXT FOR NEW SESSION / HANDOFF

Paste this section at the top of any new Claude conversation to resume work immediately without re-explaining context.

```
PROJECT: Dental AI Pipeline
INPUT: OPG X-ray
OUTPUT: Structured report — Tooth [FDI] → Caries + Bone Loss

COMPLETED:
Phase 1: Caries detection — ResNet18 binary (DONE)
Phase 2: Caries severity — ResNet18 3-class (DONE)
Phase 3: Tooth detection — YOLO (IN PROGRESS, separate)
Phase 4: Bone loss — Data exploration COMPLETE

DATASET: Tufts Dental OPG
1000 OPGs, 968 with tooth bboxes, 26005 tooth annotations
423 fully annotated images (working training pool)
Expert masks: 154 with pathology, 285 empty (normal)
Imbalance: 1.85x (moderate)
Tooth numbering: Universal 1-32 (must convert to FDI)
Bbox format: [y1, x1, y2, x2] corner points
Join key: External ID field in JSON records

ARCHITECTURE DECISION: ResNet18 binary (bone loss present/absent)
Reason: No severity GT labels in dataset
Label source: Expert mask empty/annotated + spatial IoU
Crop strategy: 1.5x width, 2.0x height (asymmetric — teeth are tall)
Normalization: ImageNet mean/std

NEXT TASK: Dataset construction
Build processed_dataset_tufts/train/0/ and /1/
Use spatial IoU crop-mask overlap for label assignment
Add +/-15% bbox jitter for YOLO noise simulation
Stratified 80/20 split

SAVED FILES IN COLAB:
df_expert.csv — 1000 expert records with stems + descriptions
df_bbox.csv — 26005 tooth bboxes with x,y,w,h
df_joined.csv — 439 expert mask records with has_bbox flag
```

8. QUICK REFERENCE

8.1 Key Code Patterns

Parse expert.json correctly

```
with open(EXPERT_JSON) as f:
data = json.load(f)
for item in data:
stem = Path(item["External ID"]).stem
desc = item["Description"]
```

Parse bbox.json — correct coordinate conversion

```
for item in bbox_data:
stem = Path(item["External ID"]).stem
for obj in item["Label"]["objects"]:
y1,x1,y2,x2 = obj["bounding box"]
x,y,w,h = x1, y1, x2-x1, y2-y1
```

Expanded crop function (asymmetric)

```
def expanded_crop(img, x, y, w, h, fw=1.5, fh=2.0):
cx, cy = x + w/2, y + h/2
nw, nh = w*fw, h*fh
x1 = max(0, int(cx - nw/2))
y1 = max(0, int(cy - nh/2))
x2 = min(img.shape[1], int(cx + nw/2))
y2 = min(img.shape[0], int(cy + nh/2))
return cv2.resize(img[y1:y2,x1:x2], (224,224))
```

Class weights for imbalanced training

```
from sklearn.utils.class_weight import compute_class_weight
weights = compute_class_weight(
"balanced", classes=np.array([0,1]),
y=train_labels)
criterion = nn.CrossEntropyLoss(
weight=torch.tensor(weights).float())
```

8.2 Dataset Path Reference (Colab)

Variable	Path
BASE	/content/drive/MyDrive/Tufts/ZIP
RADIO_DIR	BASE/Radiographs/Radiographs
BBOX_JSON	BASE/Segmentation/Segmentation/teeth_bbox.json
EXPERT_JSON	BASE/Expert/Expert/expert.json
EXPERT_MASK	BASE/Expert/Expert/mask

TEETH_DIR	BASE/Segmentation/Segmentation/teeth_mask
MAXILLO_DIR	BASE/Segmentation/Segmentation/maxillomandibular